

Reimagining Online Consent for More Responsible and Human-Centred Data Collection

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ABSTRACT

Online consent in the present day usually consists of only cookie banners, which have been heavily criticized for being annoying, deceptive, and not truly allowing for one to make an informed decision. However, we keep using cookie banners due to a lack of better alternatives.

In my PhD, I have conducted, and am currently conducting, research about issues relating to the current consent ecosystem, along with proposed solutions to improve the way online consent operates. Through my research, I hope to work towards a more responsible, transparent, and human-centred consent and data collection ecosystem.

ACM Reference Format:

Lin Kyi. 2024. Reimagining Online Consent for More Responsible and Human-Centred Data Collection. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3613905.3651129>

1 INTRODUCTION

Consent acts as a form of *moral magic*; through the action of consenting to something, we can turn otherwise impermissible acts permissible [7]. Consent is necessary and important for several daily activities, and my PhD research focuses on online consent that companies request from users before collecting and processing their data.

By consenting to online tracking, we allow companies to collect and process our data. However, the current consent mechanism, usually in the form of cookie banners, have been heavily criticized for not being transparent, allowing users to actually make informed consent decisions, and for employing deceptive practices [5, 15, 18]. Due to a lack of better alternatives, we are stuck with consent banners for the present.

My PhD research is studying consent from a variety of angles to better understand issues with the current consent ecosystem, and propose solutions that target both design and regulation. Through my research, I hope to work towards a more responsible, transparent, and human-centred consent and data collection ecosystem.

2 BACKGROUND

Under the EU General Data Protection Regulation (GDPR) in 2018, companies operating within the EU were tasked with needing to ask users for consent before collecting user data [2]. This consent must be: *freely given* (consent is given voluntarily), *specific* (consent must be sufficiently detailed), *informed* (users must be given enough information to make a choice), and *unambiguous* (users need to make a clear action specifying their consent) [2]. As such, consent notices (also known as cookie banners, privacy notices, etc.) have become the de-facto standard for collecting consent. Therefore, internet users in the European Union and beyond have become accustomed to having to deal with consent pop-ups as part of their normal internet browsing.

Despite these specifications for what consent should look like, what the law requires does not always translate perfectly to how the law is applied in practice. Multiple research studies have shown that online consent is riddled with problems, whether knowingly or not to companies. These problems related to online consent include: adding stress on users' cognitive and attentional limits, and consent notices being riddled with deceptive practice.

2.1 User Limits

The first major problem identified by researchers is that online consent notices take advantage of attentional and cognitive limits of users. Due to the ubiquity of consent notices, especially within the EU and UK, many users find them to be annoying to deal with [4, 18]. Therefore, users are less likely to meaningfully engage with them, and instead are just doing the bare minimum to get rid of the consent notices [11, 14, 18], to the detriment of users' privacy. Consent notices, while developed with good intentions in mind, do not translate these good intentions to users in practice if they burden users' cognitive and attentional capacities.

Consent notices are a choice proliferation problem for users, where there are too many choices present, making users feel overwhelmed [14], while also not providing enough choices, such as making it difficult to decline tracking [5, 18]. Therefore, consent notices present a conundrum to users, where they make many users feel overwhelmed and resigned to share their data with organizations, despite the "choice" that consent notices give them [11, 13]. Overall, these practices have made users distrustful towards consent notices, while not making them more informed about how their data is being used [10, 11].

2.2 Deceptive Practices

Whether knowingly or not [4], most consent notices make use of deceptive practices, such as deceptive design (dark patterns), and deceptive language to encourage users to consent to cookies [15, 16].

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CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
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ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3651129>

A notable deceptive practice is making it easy for users to consent to cookies while making it difficult to decline cookies [14, 18], and researchers have also noted that the placement and design of a consent notice impacts users' consent choices [5, 18]. Additionally, another deceptive practice is deceptive language, such as when the benefits of consenting to cookies are outlined, while the negatives are often not mentioned by companies [11, 16].

Due to the ubiquity of consent notices, in addition to the well-known issues consent notices present, more research has to look into how we can improve, and perhaps replace, the online consent ecosystem. But the question remains: how can organizations collect consent from users, while not adding additional cognitive burden or using deceptive practices?

3 RESEARCH GOALS

Through my PhD research and dissertation, I hope to work towards reimagining the online consent ecosystem. Change, especially for a large and multi-stakeholder online consent ecosystem, takes time, and hopefully by building upon previous work, online consent can become more human-centric and be more responsible when collecting user data.

The majority of computer science studies studying online consent have only looked at the consent notices themselves, such as studying the usability of them [5, 14, 18], the direct impacts on users [11, 14], and measurement studies of consent notices [6, 15].

In my PhD, I am taking a broader look at the human aspects of online consent. While consent notices have become the de-facto standard for collecting consent, there are other peripheral aspects that must be studied to better understand how we can work towards a more human-centred and responsible consent ecosystem.

Questions I have investigated, or hope to investigate, in my PhD include:

- (1) What is the role of GDPR's legitimate interest, a legal ground that allows companies to collect and process user data *without* user consent (GDPR Article 6 (1)(f)), in the consent ecosystem, and how might it be potentially exploited by companies as a method for collecting more user data?
- (2) What are user perceptions of the data collection purposes they are consenting to, and do consent notices actually do a good job at informing users?
- (3) In situations where consent is not possible, such as in the case of generative AI scraping data to train its models, what are some guidelines to inform more ethical data collection practices from involved stakeholders?
- (4) Based on the findings from these previous questions, how can we revamp the online consent ecosystem in a way that is more privacy-preserving, responsible, and generalizable to different contexts, such as new AI advancements?

Online consent consists of the consent notices themselves, which have received the most attention, in addition to peripheral aspects, which have not received as much attention, but still have a large impact on the consent ecosystem. By investigating the broader ecosystem consisting of consent notices in addition to the peripheral aspects, I hope to arrive at a more holistic conclusion for how online consent can be revamped, or even potentially replaced with a more usable and responsible alternative.

4 RESEARCH METHODS AND APPROACH

With a background in psychology, I have a keen interest in user study research methods, ranging from qualitative to quantitative methods, and combining qualitative and quantitative methods.

For my qualitative methods, I have conducted a content analysis of output from a web crawl, content analysis of social media posts, and have conducted and analyzed semi-structured interviews and co-designs with participants. For quantitative methods, I have conducted and analyzed user surveys and experiments using inferential statistical methods.

I prefer qualitative methods when it comes to understanding a phenomenon deeply, especially if it is in an understudied research topic, and prefer quantitative methods when I hope we can generalize our findings to the wider population, and to see when different conditions are statistically different compared to than others.

As the topic of my dissertation spans across multiple disciplines, it is important to also collaborate with an interdisciplinary team of scholars. I collaborate with computer scientists from more different areas of expertise (security and privacy, information retrieval, HCI), legal scholars, and scholars with a behavioural science background. From these collaborations, I have gained valuable insights on how to ensure our research is relevant for these disciplines beyond HCI, and have been exposed to research literature from these disciplines to apply in our studies.

5 RESULTS TO DATE

In this section, I will describe the results of two papers that have been published, or accepted for publication related to my dissertation.

For our CHI 2023 paper, "Investigating Deceptive Design in GDPR's Legitimate Interest" [12], we took a look at how legitimate interests, one of the most flexible and vague legal grounds for data collection under the GDPR that legal scholars have speculated may be prone to abuse [3, 9], is being used in practice. We measured its prevalence in consent notices, identified the deceptive designs being used when it is applied by creating a taxonomy of legitimate interest-specific deceptive designs, and ran a user survey to see how users evaluated these practices.

Our findings from the web crawl analysis were that legitimate interests have the potential to be abused by organizations as a loophole for collecting data without consent. This is due to the several deceptive practices we identified in our analysis, such as how legitimate interests are more difficult to reject tracking to compared to consent. From our survey, we found that the current uses of legitimate interests do not line up with user expectations, and that more consideration needs to be paid to align user interests with business interests when it comes to applying legitimate interests. Based on our findings, we suggested design and legal recommendations to ensure more responsible uses of legitimate interests in the future.

The second paper in my PhD dissertation is "It doesn't tell me anything about how my data is used": User Perceptions of Data Collection Purposes", which was recently accepted for publication at CHI 2024 [13]. In this paper, we studied six common data collection purposes (*Strictly Necessary, Statistics and Analytics, Performance and Functionality, Marketing and Advertising, Personalized Advertising, and Personalized Content*) to identify elements of an effective

purpose name and description in consent notices presented to users to improve the *informed* aspect of informed consent.

Some data collection purpose names, such as “*Statistics and Analytics*” and “*Performance and Functionality*”, were unclear to participants about what they do with user data, and therefore may require revising to be more transparent. For purpose descriptions, we found that the text within consent notices does little to actually inform users about how their data is being used by organizations. Based on our findings, we suggested new potential purpose names that fit users’ perceptions of what these purposes do, and discussed the types of information users want to learn about in consent notices to improve purpose descriptions.

6 EXPECTED NEXT STEPS

I have completed two studies outlined in the previous section, and have two studies left to complete in my dissertation. I hope to submit my dissertation by late 2025, and defend my PhD in early 2026.

The first project I am currently working on (the third project in my dissertation) is about investigating the impacts of generative AI on creative workers. Since the rise in generative AI’s popularity, the creative community has spoken out about the economic losses, copyright issues, and career-based harms they have encountered [8]. Through our study, we hope to provide guidelines regarding the responsible collection of data for training generative AI and better promote the rights of creative workers.

The next project I am planning to work on is to better incorporate principles of informed consent from different disciplines into online consent. Online consent is a topic of discussion not only in HCI, but also in law [16], machine learning [17], philosophy [1], among other disciplines. I hope to understand and incorporate consent perspectives from these different disciplines to propose potential alternatives to the consent ecosystem, ideally alternatives that understands the needs and challenges of users, and are generalizable to new advancements in AI.

7 LONG-TERM GOALS

My dissertation will be written based on these completed and expected projects. Currently, two of those projects are complete, one is underway, and I have begun the early stages of my final dissertation project. The plan is to submit and defend my dissertation by 2026. In addition to my own dissertation, I am exploring other research topics to expand my skills and knowledge of the HCI field, such as collaborating on a project about digital well-being, and leading another project about the role online forums play in legal advice.

My long-term goals are to remain in research, and hope to remain in the CHI community after my PhD, whether in academia or in industry. I also hope to stay in my general area of research (privacy and responsible technology), but hope to broaden my research focus once my PhD is done, and explore adjacent research communities such as the machine learning ethics community.

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